ABSTRACT OF THE DISCLOSURE

A system and method for encoding data on a high-cardinality parallel communication channel by creating logical groupings of physical channels, termed "n-grams." The method comprises: defining a group of 'n' adjacent physical channels on a parallel interface; treating the combined state-space of the 'n' channels as a single logical channel with a vastly expanded number of states; and mapping a large data block to a single state of this logical n-gram channel. For example, by treating two adjacent 24-bit pixels ("bigram") as a single logical channel, a 48-bit (6-byte) data block can be encoded in a single transmission step. This method provides an exponential increase in data density and enables a form of over-the-channel data compression.